ASI Boston Hackathon 2026 - Guide

Welcome! This page is your reference for the day; challenge details, resources, schedule, and logistics. The Claude API key and data bundle setup guides are in the toggles below.

ℹ️ **At a glance**

- 📅 Saturday, May 30, 2026
- 🕘 9:00 AM – 5:00 PM ET
- 🆘 Need help? ASI support staff are on-site — flag any of us down

The Challenge

Given a day of US flight plans, weather, and airspace information, build something that makes the system better.

You can build:

- A routing optimization algorithm
- A decision-support tool
- A 4D data visualization
- An interactive simulation or game
- Anything else you can demo and explain why it's interesting

Judging Criteria

- **Approach & technical merit**
- **Insight & problem understanding**
- **Communication & clarity**
- **Creativity**

Your Resources

You'll have two resources to work with for this challenge:

- A curated open-source data bundle covering US flights, airspace sectors, and weather — everything you need to start building immediately, no external data sources required (though you're welcome to add your own).
- Claude API keys — one dedicated API key per person, valid for the day, usable with the Claude Code CLI and/or any editors that accept an Anthropic API key, like Cursor. Using the Agent SDK and API directly is also permitted.

The toggles below walk through how to access each: where to download the data bundle, and how to install, configure, and verify your Claude API key.

▼ Claude API keys

What you get

Each person gets a dedicated Anthropic API key valid for the day.

You can use it via the Claude Code CLI and/or any editors that accept an Anthropic API key, like Cursor. Using the Agent SDK and API directly is also permitted.

Note: API key auth is not supported in Claude Desktop or Claude Web.

Claude Code CLI Installation

macOS, Linux, WSL:

```
curl -fsSL https://claude.ai/install.sh | bash
```

Windows PowerShell:

```
irm https://claude.ai/install.ps1 | iex
```

More info here: Claude Code Docs Overview - Claude Code Docs

How to access your key

You'll receive a Bitwarden Send link by email the morning of the hackathon. The link can only be opened with a password and expires at the end of the day.

Need help getting set up? ASI support staff is on-site for assistance, flag one of us down.

Ground rules

- Don't share your key with others. Everyone gets their own key.

Using your key in the Claude Code CLI

Set the API key as an environment variable in your terminal.

macOS / Linux:

```
export ANTHROPIC_API_KEY="sk-ant-..."
```

Windows (PowerShell):

```
$env:ANTHROPIC_API_KEY = "sk-ant-..."
```

Note: this only persists for the current terminal session. For every new terminal window you will need to set it again, or save it in a config file like `~/.bashrc`.

Verifying your key is being used

First, run the command `claude` in your terminal.

If you are setting up Claude Code for the first time: after answering an initial color theme prompt, you should see a message that says `Detected a custom API key in your environment`. Select `Yes`. If you by mistake select `No`, delete your `~/.claude.json` file and try again.

If you are already a Claude Code user: your API key should take precedence over most login methods. To confirm, once in a Claude session, run the `/status` command. You should see `API key: ANTHROPIC_API_KEY`. If not, run `/config` and ensure `Use custom API key` is set to `true`.

Using your key in Cursor

Cursor Documentation Bring your own API key | Cursor Docs

Where to learn more

- https://docs.claude.com/en/api
- https://code.claude.com/docs
- https://cursor.com/en-US/docs

▼ Data

What's in the bundle

A snapshot of US air traffic, the airspace it flies through, and the weather it deals with.

- All timestamps are **UTC**
- Coverage is **continental US (CONUS)**

Included data:

We have aggregated some open source data from interesting days to help you get started.

- **Flights & planned routes**: scheduled US flights with planned lat/lon waypoints, cruise altitude, and cruise speed
- **Airspace sectors**: synthetic ATC sectors partitioning CONUS airspace into two altitude bands, each with a capacity
  - Gathered from https://www.weather.gov/zse/ (capacities estimated by size)
- **Weather forecasts**: HRRR-derived precipitation intensity (`refc`) and storm-top altitude (`retop`) as gridded arrays, in 15-minute strips extending ~18 hours forward
  - Gathered from https://home.chpc.utah.edu/~u0553130/Brian_Blaylock/cgi-bin/hrrr_download.cgi

How it’s organized

- Top-level `documentation/` folder (with a `FILE_FORMAT.md` per data type)
- Shared `sectors.geojson.gz`
- 11 scenario directories named `asked_at_<YYYY-MM-DD>T<HH:MM:SS>Z/`, each containing:
  - `routes.json.gz`
  - `wx/` folder with `refc/` and `retop/` `.npz` files

Scenarios span May 2025 to April 2026 and are self-contained.

Where to learn more

Each data type has a `FILE_FORMAT.md` in `documentation/` with:

- schema
- units & conventions
- short Python reading/plotting examples

Read these first.

Bringing your own data

You're encouraged to enrich this bundle with open-source data of your choosing to strengthen your solution. Pick whatever fits your idea, just mind each source's license and cite what you use.

How to access it

Please use the pre-signed S3 URL below to access the bundle as a single archive.

📈 Presigned URL

Why This Matters

Today you will have the opportunity to solve the critical problems affecting the National Airspace System. Every day, the Federal Aviation Administration handles about 45,000 flights and moves over 2.5 million passengers. Technical issues, weather and congestion cause 90 million minutes of delay per year. Improvements to the system are needed to keep air traffic moving smoothly and on time.

Every day, ASI helps airlines navigate exactly these issues. For this hackathon, we're handing you a slice of the same problem.

On December 22, 2022, a single winter storm cost US airlines $150M and stranded a million passengers over Christmas. Decisions about which flights to delay, reroute, or cancel ripple across the entire network. These decisions have to be made in minutes.

Schedule

Schedule

- **8:45–9:15**: Check-in, self-organize into teams (max 4 per team)
- **9:15–9:30**: Welcome, problem briefing, rules of the road, scoring criteria, judges intro
- **9:30–9:45**: Final setup
- **9:45**: Hack begins
- **2:30–3:00**: Finalize your demo
- **3:00–3:30**: Demo showcase (judges visit each table)
- **3:30–3:45**: Judges select top 5
- **3:45–4:15**: Top 5 present to the room (3 min demo + 1 judge question each)
- **4:15–4:30**: Judges deliberate
- **4:30**: Winners announced
- **4:30–5:00**: Mingle
- **5:00**: End

Demo, Judging & Prizes

Format

- **3:00–3:30: Showcase at your table.** Every team demos in place. ASI judges rotate table-to-table so each team gets face time and a chance to walk through their solution.
- **3:45–4:15: Top 5 present to the room.** Judges pick five teams during the showcase. Each gets a 3-minute live demo plus a question from a judge.

What to Prepare